ARCHITECTURE AND DEPLOYMENT

Simple to run. Yours to host.

React console + Go API + PostgreSQL. Optional native desktop launchers wrap the same stack for local evaluation.

Runtime pieces

Console	React 19 + Vite SPA. Desktop / install builds serve it same-origin from the API (`UI_DIR`).
API	Go (chi, pgx). 84-entry feature catalogue (73 modules + 11 AI), batch/lab/QA modules, market stubs, hash-chained audit, background scheduler and connector delivery worker in one process.
Database	PostgreSQL with 29 forward SQL migrations applied at start-up; opt-in down scripts for the newest three only (back up first).
Desktop	`pharma-desktop` / PharmaQMS.app / PharmaQMS.exe — start Postgres if needed, run API+UI, open the browser.

Deploy paths

Desktop

`make desktop` · `make desktop-macos` · `make desktop-windows`. Best for an afternoon try.

Compose

Podman/Docker Compose for Postgres + services on Mac / Linux / WSL.

FluxVM

macOS FluxVM / lima path for a contained plant-like stack.

k3s

Helm chart and raw manifests for a small cluster pilot; one API replica by default.

Windows

Docker Desktop compose script, or WSL2 running the Linux compose path.

Install bundle

The public installer scripts (`get.sh`, `get.ps1`) expect a bundle and checksums on the site Releases page. **None is published yet**; evaluation bundles are available on request. No application source on the public site.

Before anyone else can reach it

Shipped deployments start in dev mode (identity headers trusted, default admin). Enable strict authentication, mark the environment production, set a one-time bootstrap password, terminate TLS and overwrite forwarded-for headers at a proxy. Other switches, at a high level: allowed origins, a connector secret key, passkey / OIDC / SAML settings, an opt-in for private connector targets, and an AI mode switch. Change the demo database credentials in the shipped files.

Default local endpoints (desktop / install)

Health	`/healthz` on the local address, port 8080
API + UI	One local port, 8080 (the API serves the built UI)
Dev split	API :8080 · Vite UI :5173

Pilot only. Single node (multi-replica untested), not production-hardened, not a validated GxP hosting claim.